

Sky Bound, Yet Online

"I'm travelling with limited access to Internet" might soon be a redundant out-of-office mail responder!



JANANI GOPALAKRISHNAN VIKRAM

We can access the Internet about 11 km (36,000 feet) above the ground at \$10 per flight, thanks to in-flight Wi-Fi. But, for users, it is still too expensive and slow. With hardly any bandwidth limitation and the ability to transmit data to flights over land and water, satellites appear to be a more attractive solution for in-flight Internet now.

A decade-old love story

In-flight Internet is something passengers have been wanting badly for a long time and operators have been unable to fulfil with full features.

Way back in 2003, Boeing experimented with Connexion—an in-flight Internet technology that used high-speed signals from geostationary satellites and special receivers fitted on the aircraft. Although many partners showed interest initially, the momentum slumped following the 09/11 attacks in the US, which resulted in a lot of security fears and funding issues.

A few years later, Aircell started offering an air-to-ground (ATG) Internet service, which it later re-branded as Gogo Inflight Internet. Gogo has been operating since 2008, and is now a popular service adopted by many airlines in USA including American Airlines, US Airways, Delta Airlines, Virgin America and Frontier Airlines.

Go, go, data on the flight

Gogo transmits data to planes via cell towers on the ground, making it very cheap and easy to implement compared to satellite-based systems like that tried by Boeing. Their ATG system is based on a 3G wireless technology that utilises EV-DO Rev. A. Gogo owns more than a hundred towers, which are actually cell-phone towers outfitted to point their signals at the sky rather than along the ground. The aircraft picks up the signal through a receiver and distributes it throughout the cabin using a Wi-Fi system.

Gogo charges between \$5 and \$10 depending on the duration of the flight. If you use simple e-mail, chat,

Twitter or such services, the speed is reasonably fine. That said, recent surveys by consulting firms show that only 8 per cent of flyers in the US use the Internet on-board. As more people start using the Internet on the fly, it is expected that the speed of Gogo's service will deteriorate badly.

For a short spell, when Google decided to sponsor Gogo on some flights, the usage rates increased quickly and the speeds apparently became terribly slow. This incident demonstrated the effect of high traffic on Gogo-like services.

European services based on similar technology also face such problems. During a recent trip, a director at UK-based telecom consulting firm Machina Research found the in-flight Internet to be so slow that he later remarked "My experience was on Norwegian and it was free. And it would have to be! I probably wouldn't have paid for it."

Satellite on the rise

The inability of Gogo and such services to support a large number of users and